

Alleviate

Movement Made Seamless



CP02 Group 3:

Chan Si En, Chua Heng Li, Lim Tong En Saffron, Tan De Jing, Victoria Syn Hui En



Solutions Explored

Concept	Target Users	Problem	Proposed Solution
Smart Walker	Elderly (Dementia)	Wandering, getting lost	Vibration-based navigation + GPS tracking
Shower Assist System	Elderly / Mobility-impaired	Difficulty bending to clean body	Adjustable chair / Rotating scrub bar
Anti-slip Smart Mat	Disabled individuals (Shower)	Slipping and falling	Force-sensitive mat that detects and alerts instability
Bus Assist Device	Visually impaired	Cannot identify correct bus	Bluetooth device with tactile numbers + vibration/audio alerts
Sit-to-Stand Aid (Alleviate)	Elderly	Difficulty getting out of bed	Mechanical rail + sliding mat that lifts and guides user to standing
Item Finder System	Elderly (Early Dementia)	Misplacing daily items	Bluetooth tags + voice assistant + buzzer/LED
Physiotherapy Glove/Game	Stroke patients	Low motivation for rehab	Sensor glove connected to gamified therapy platform

Pugh Chart

Concept	Feasibility	Tech Complexibility	Impact	Innovation	Simulation Ease	Total
Smart Walker	+1	0	+1	0	+1	+3
Shower Assist System	-1	-1	+1	0	+1	0
Anti-slip Smart Mat	0	+1	0	+1	+1	+3
Bus Assist Device	+1	0	+1	+1	+1	+4
Sit-to-Stand Aid (BED concept)	+1	+1	+1	+1	+1	+5
Item Finder System	+1	0	0	0	+1	+2
Physiotherapy Glove/Game	+1	+1	+1	0	+1	+4



Rationale

for choosing Sit-to-Stand

27.6%
of elderly struggle with bed exits.¹

38%
of caregivers report physical stress.²



- | Restoring user's independence/autonomy
- | Mitigates physical burden on caregivers
- | Healthcare efficiency

¹ Commons, J., Mount, J., Kresge, L., Klaus, G., Mann, L., & Palomba, C. (2006). Movement patterns used by the elderly when getting out of bed.

² Darragh, A. R., Sommerich, C. M., Lavender, S. A., Tanner, K. J., Vogel, K., & Campo, M. (2015). Musculoskeletal Discomfort, Physical Demand and Caregiving Activities in Informal Caregivers. *Journal of Applied Gerontology: The Official Journal of the Southern Gerontological Society*, 34(6), 734–760. <https://doi.org/10.1177/0733464813496464>



Problem Statement

How might we enable elderly with limited strength to safely transition between sitting on a bed and standing independently?



Market Research



Patient Hoists

- Requires caregivers to handle
- Takes time to position user on mat



Power Sit-to-Stand Lift

- Sudden jerk when rising
- Bulky equipment



Stand-Assist Chair

- Not applicable for the bed



Standing Frame

- Requires strength from user's hands

Interviews (1)



Tan Boon Seng, 87

“Get out of bed hard you know! Need to **turn here turn there.** That time I hospital then I kenna lift in air. Walao, bones falling apart already but they still lift me up. Siao!”

Lee Ling Hua, 84, Mr Tan’s Wife and Caregiver

“Yah now he stay at home and when he want to go out watch TV, he ask **me to help him up.** Bopian I’m his wife mah, end up my **back ache everywhere ache.** But hospital bed so **expensive,** who can buy in this economy? A few more years die already, buy for what?”



Thomas Liew, 19, NSF

“Erm sometimes I need help my grandma up also ah. But I Commando so no problem. But if I not at home then a bit **worrying.**”



Interviews (2)

Tan Boon Seng, 87

“Huh?! So it help me to stand up from below (at the end of) the bed ah? Eh ho seh (good) like that I no need turn. That day I try getting up without bothering Hwa then I fall down you know!”
“Can let me try if you build?”



Lee Ling Hua, 84, Mr Tan's Wife and Caregiver

“Ah I see, so this one my husband **can use himself**,
no need give me back pain?”
“But this one will need his **weight** or not? He quite
fat.”



Thomas Liew, 19, NSF

“Err I feel like my grandma would use this.
TBH **I would use also** ah. Hard to get out of
bed in camp for reveille sia hahah.”



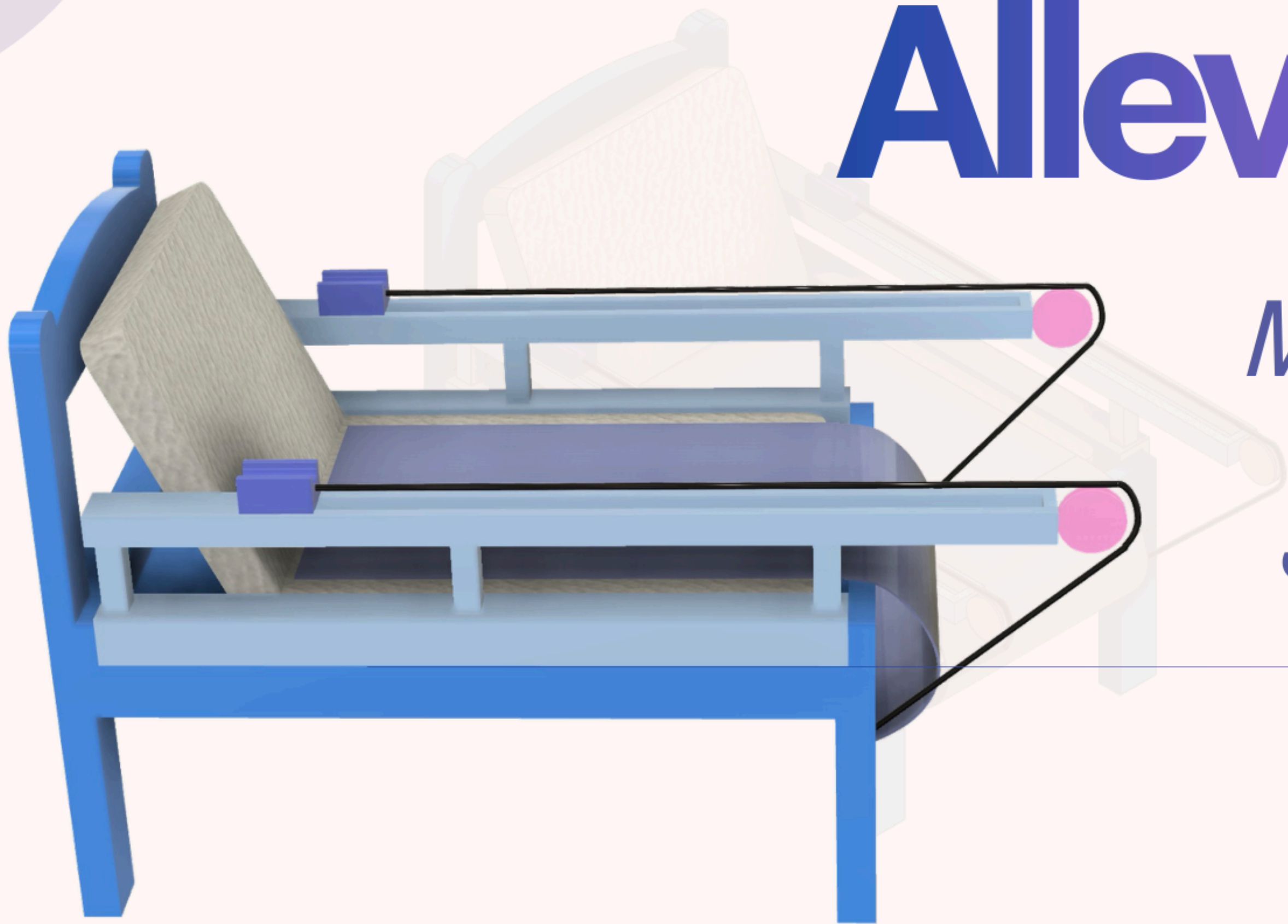


Our solution...

Alleviate



*Movement
Made
Seamless*

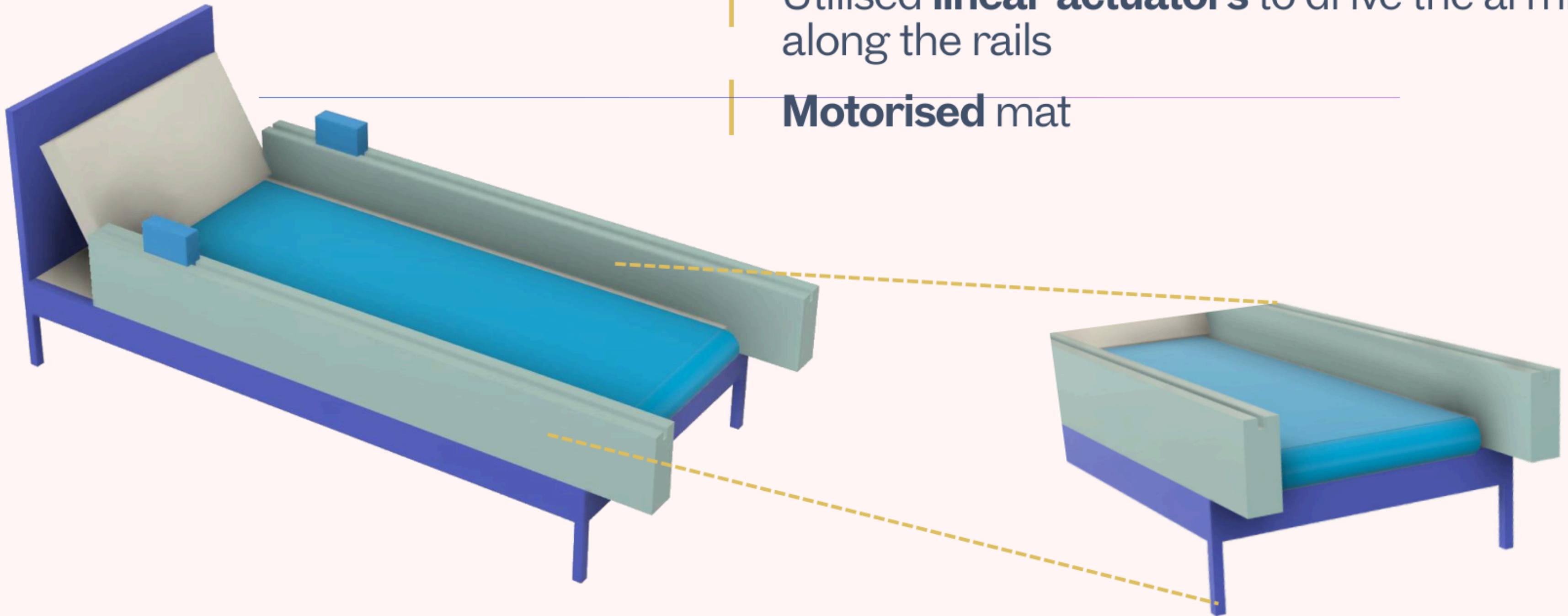


Initial Design



Utilised **linear actuators** to drive the armrests along the rails

Motorised mat

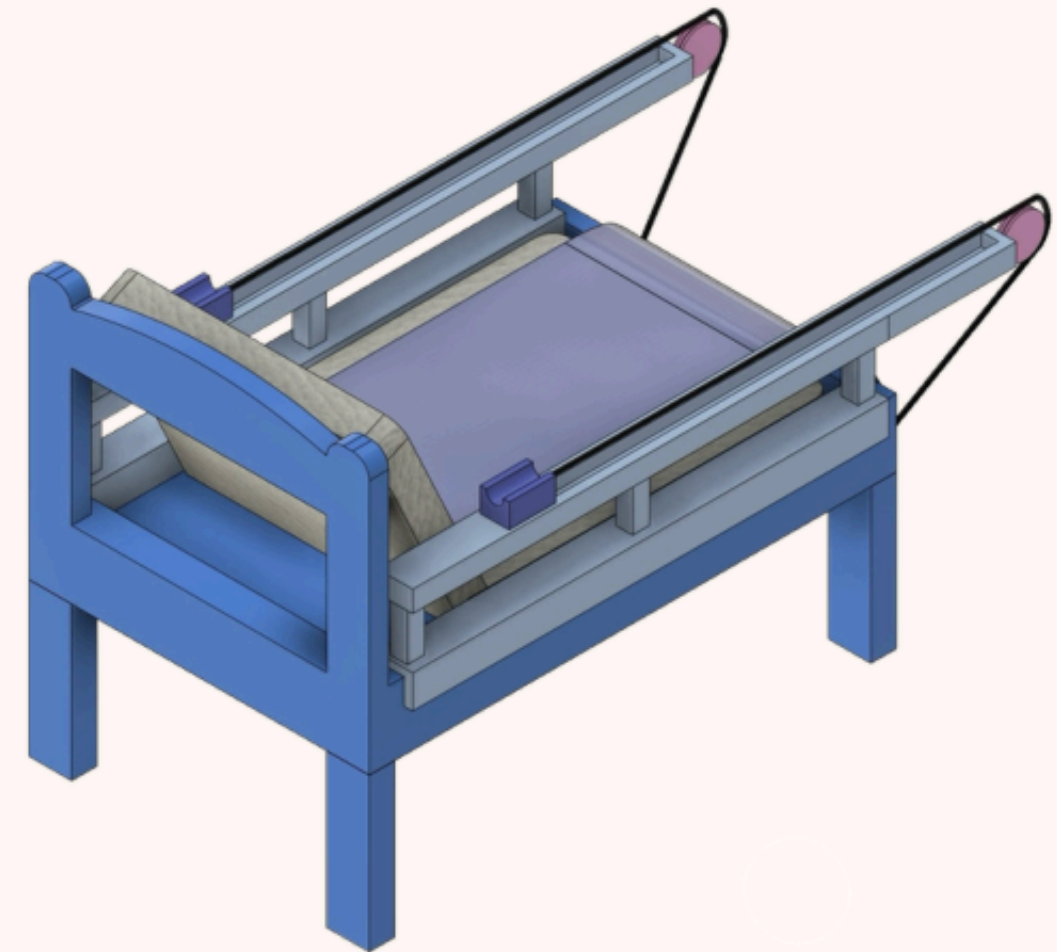
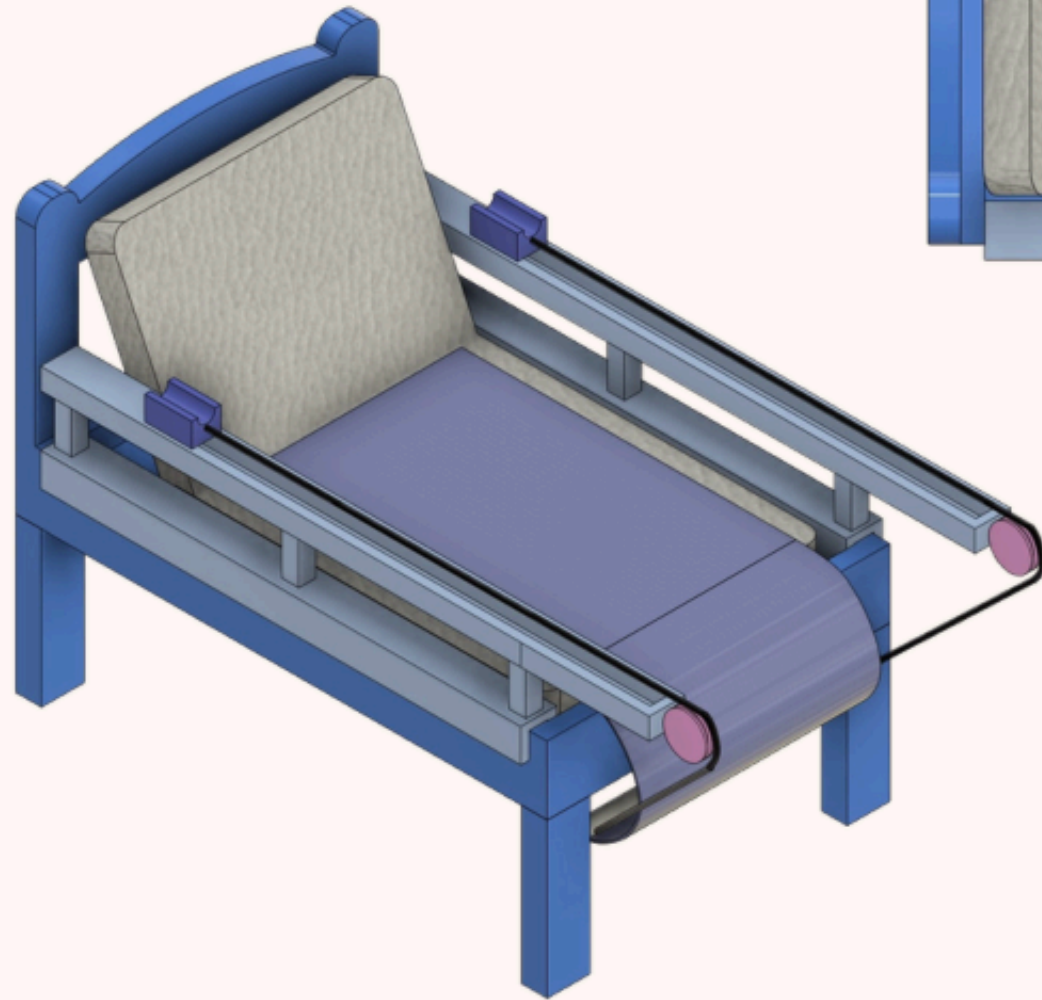
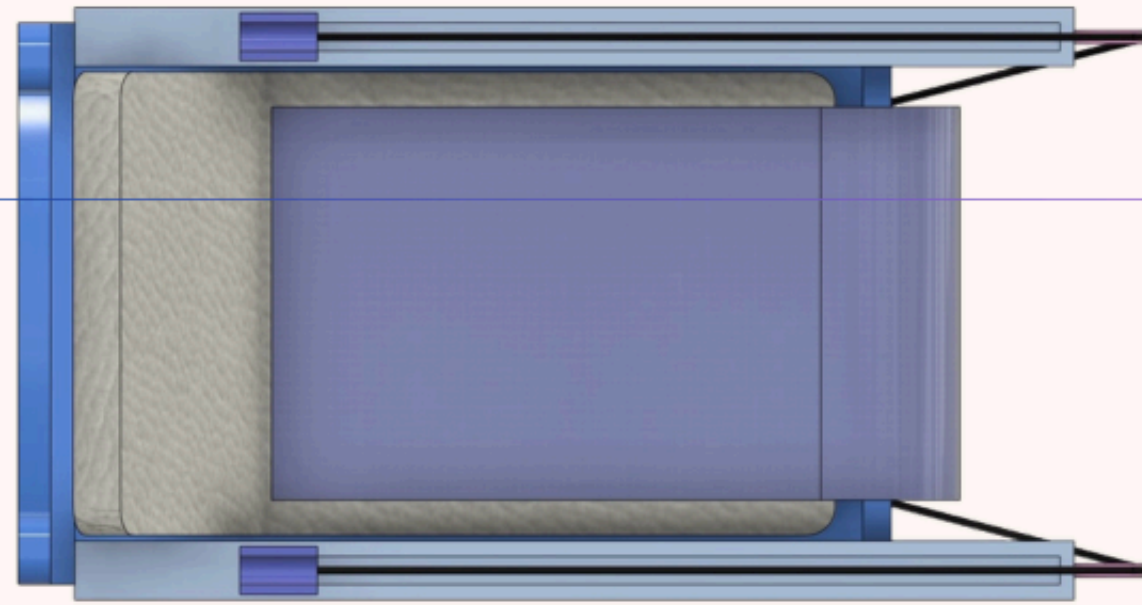
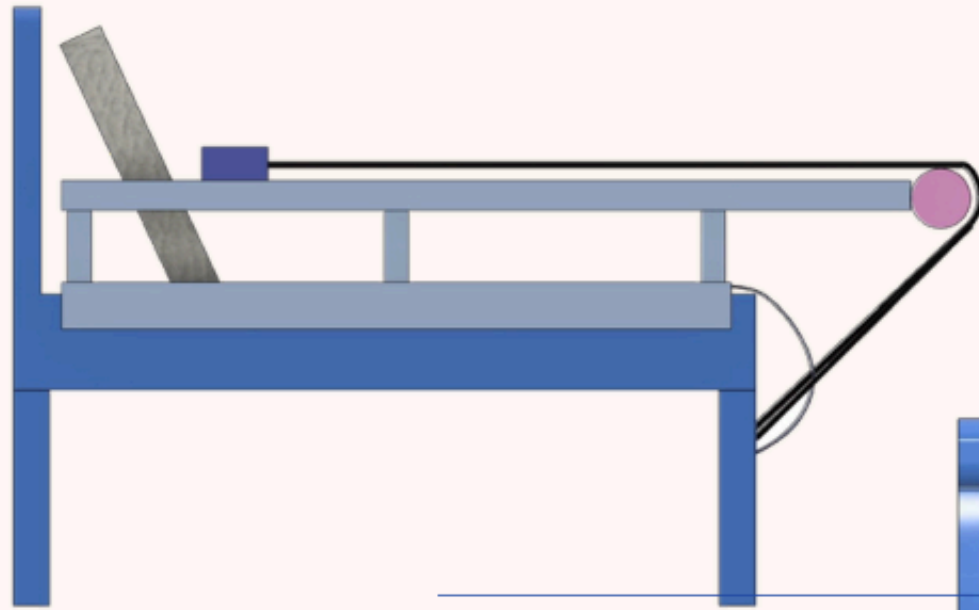


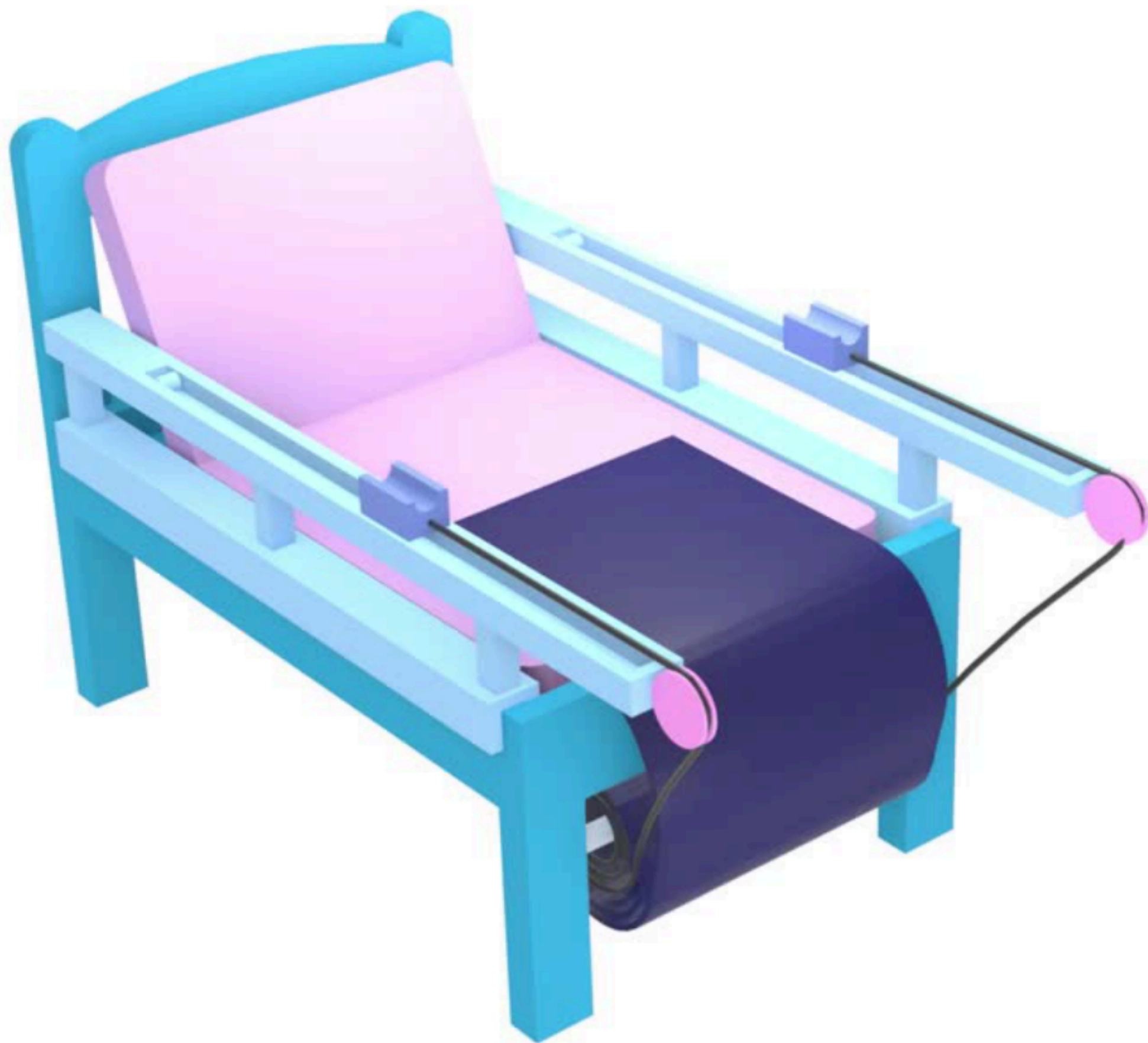
Final Design



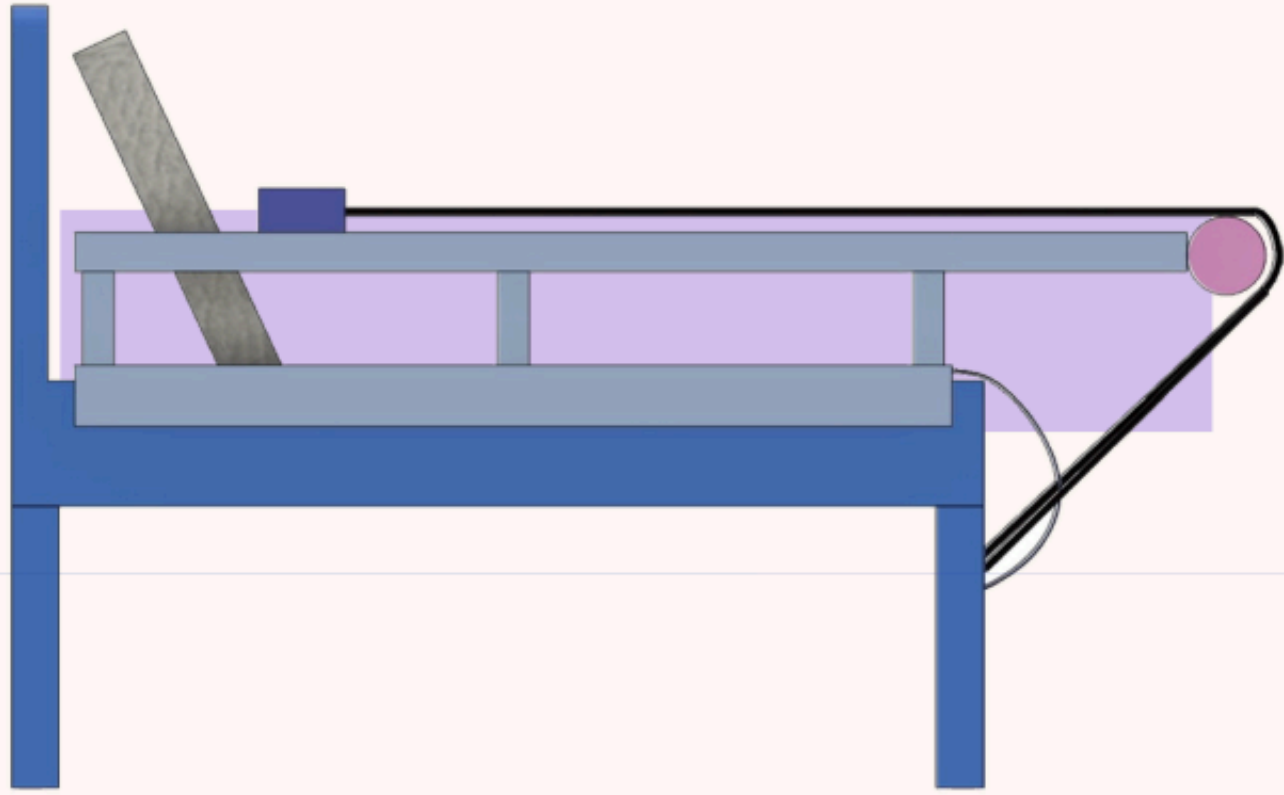
Centralised motorisation
with a pulley system

Mechanical
synchronisation using a
shaft





Structures & Materials



Stress & Beam Analysis: Determines the material and thickness of the rails

Material Analysis: Narrows down the materials for the mat

Factor of Safety: FOS on the shaft + pulley system



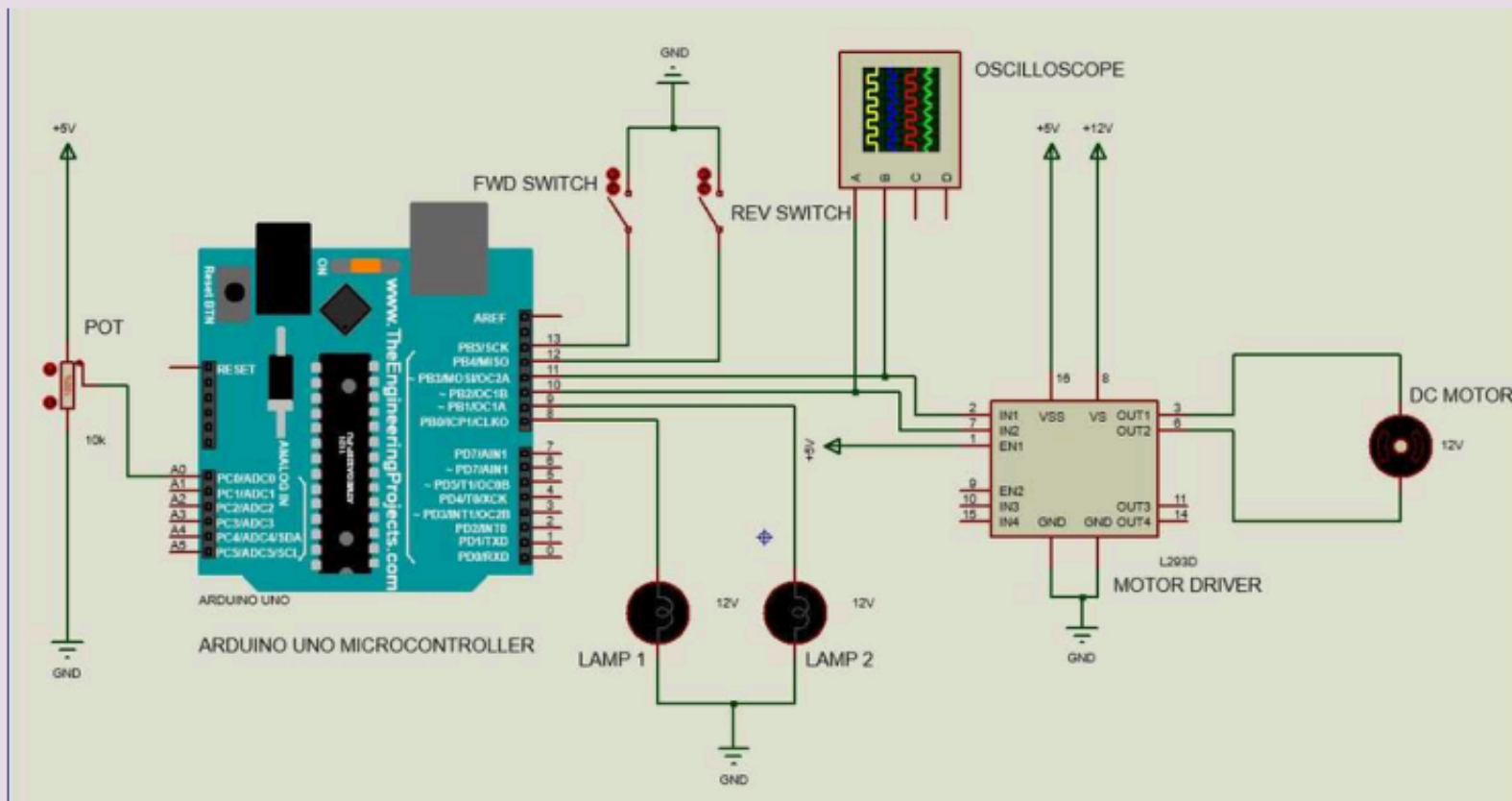
Circuits



The system is powered by an embedded control architecture designed to ensure safe, precise, and user-friendly operation.

Motor Control System

A DC motor driver circuit, controlled by an Arduino microcontroller, regulates both the **direction** and **speed** of the motor.



Safety Mechanism (Limit Switches)

Limit switches are installed at the ends of the rail travel to stop the motor once the system reaches its physical limits.

This ensures:

- **No overextension** of mechanical components
- Fully **automatic stopping** without user intervention



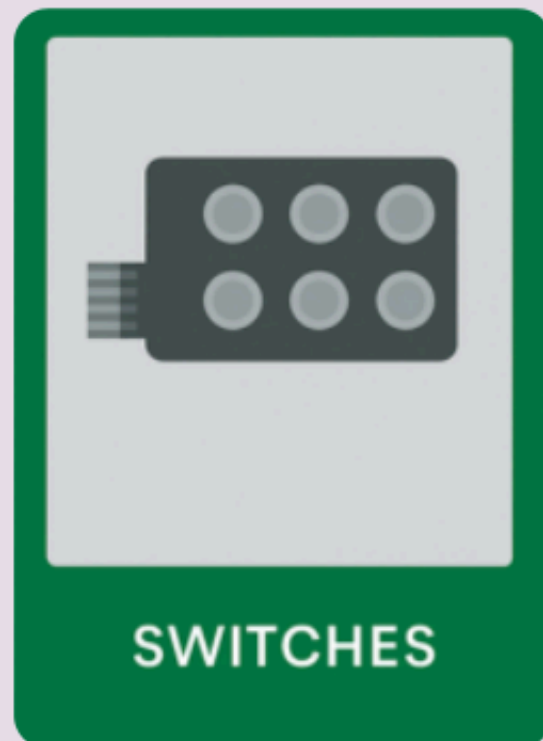
Circuits



User Interface

System is operated through a simple button-based interface, allowing users to **operate it easily**:

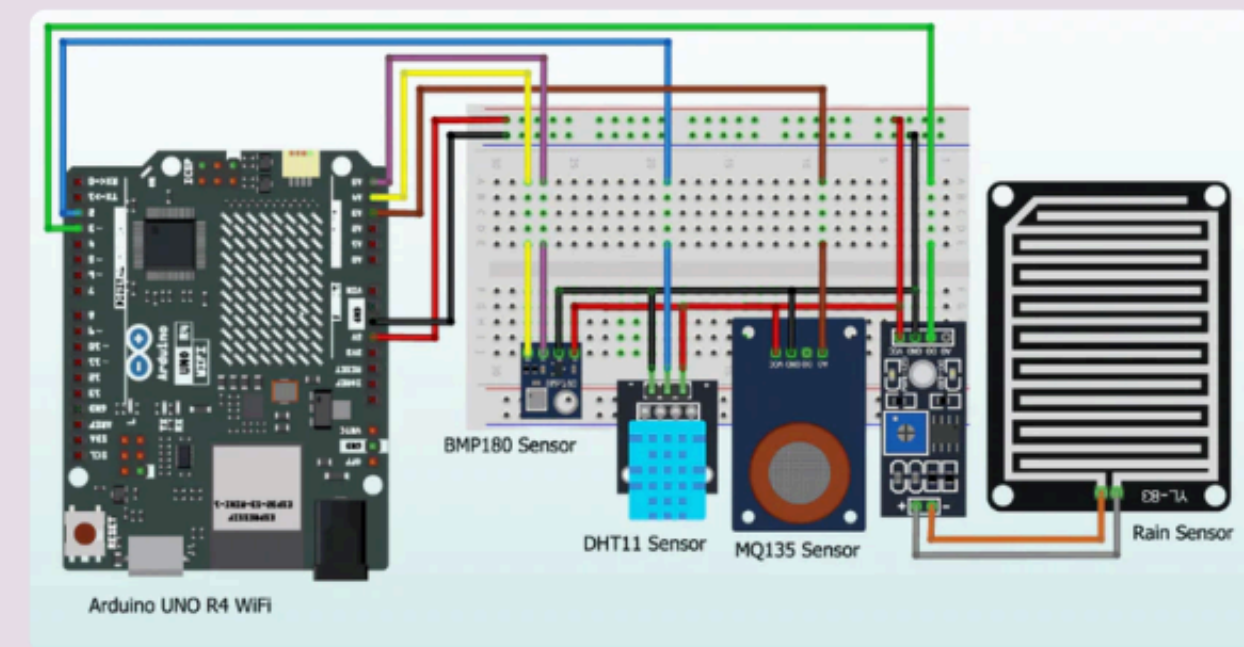
- Start / Stop button
- Forward / Reverse controls



Real-Time Position & Motion Coordination

Incorporates timing and sensing to coordinate movement between the armrests and mat.

Using integrated circuit components (timers, sensors, feedback signals), the Arduino tracks the position and motion state of the system, ensuring that the **correct motor actions** are **triggered at each stage of the transfer**





Model the Relationship between Shaft Rotation Angle and Carriage Displacement

Define initial and boundary conditions, Start position and end position at physical limits of the track, Shaft diameter.

Assuming:

Wire cannot be extended

Wire wraps around shaft uniformly with negligible thickness

Wire doesn't slip

Numerically solve the relationship using Python

Visualise the relationships for both theoretical and actual displacements.

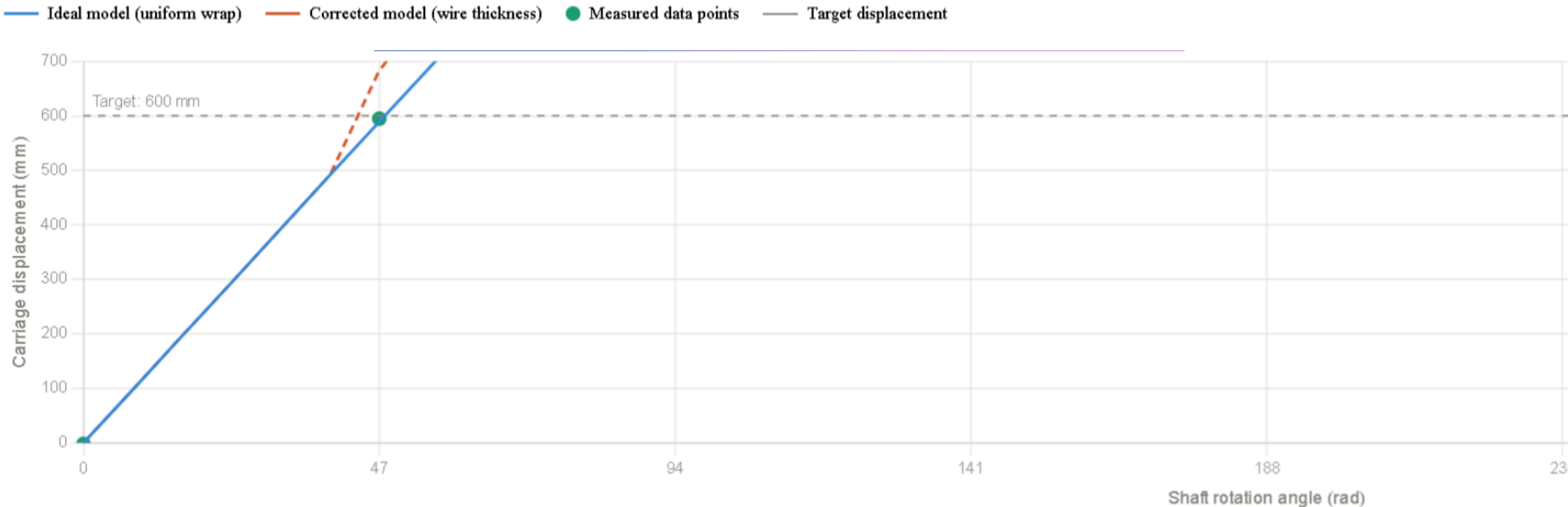
Shaft rotation required informs us on the **motor specification** needed



CaDDE

Model 1 — kinematic model

Carriage displacement vs. shaft rotation angle





CaDDE

Model the motor torque required based on user weight and friction/mat friction coefficient

Use range of weights and friction coefficients, and compute the range of **required motor torque** such that the motor does not fail in reality.

Choose motor specification that works for the **majority** of users.

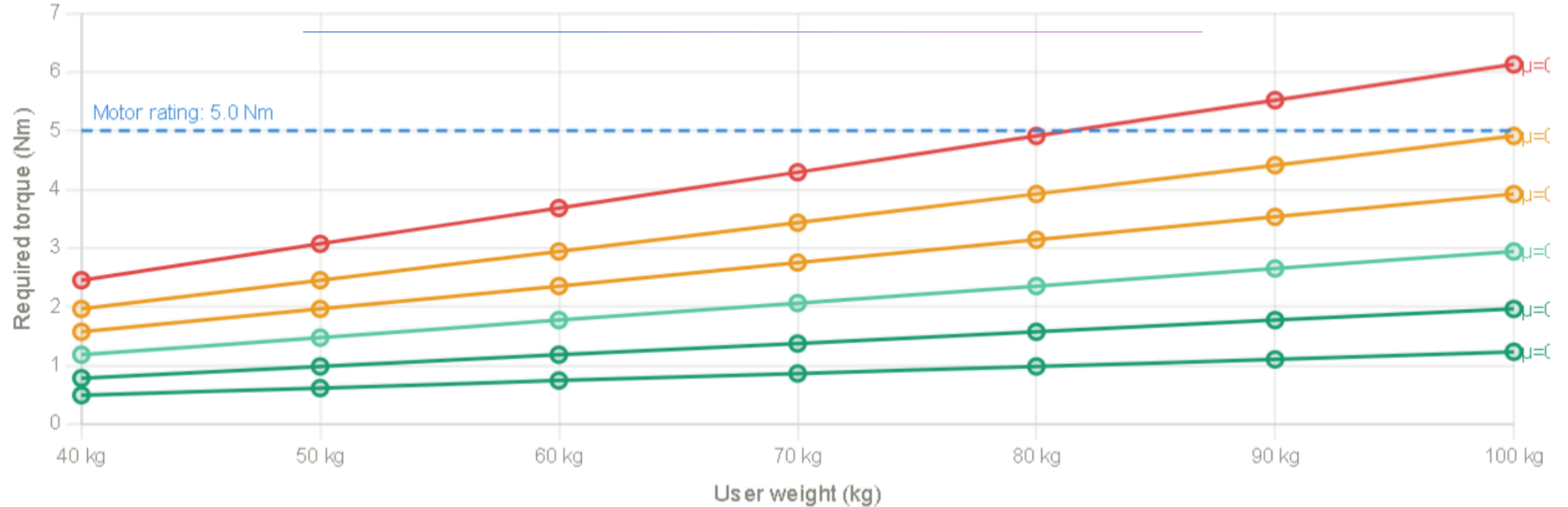
Choose torque for motor that is at least **1.5-2 times** the worst case scenario

CaDDE

Model 2 — torque model

Required motor torque across user weight and friction coefficient

Safe zone Margin zone Exceeds motor rating



$$T = \mu \cdot m \cdot g \cdot (d/2) \cdot \text{safety factor} (\times 2)$$



Thank you

